

Design of Humanoid Robot Stable Walking Using Inverse Kinematics and Zero Moment Point

Ario Sunar Baskoro, Mark Gabriel Priyono

Department of Mechanical Engineering, Faculty of Engineering
University of Indonesia
Indonesia

Email: Ario@eng.ui.ac.id

Abstract- Main problem of humanoid locomotion is walking stability. Goal of this research is designing and standardize the stability of robot. Commonly, research of humanoid robot uses the method of static walking or dynamic walking for its stability. The usage of static walking on robots emphasize on stability pose while dynamic walking emphasizes on its mobility efficiency. The combination of both walking methods can achieve a stable and efficient humanoid robot movement, in which applies the rule of Zero Moment Point and Inverse Kinematic. Optimize centre of mass from humanoid robot with selection of combination material. This research will be done by simulation and real examiner to get a stability margin and also velocity of robot to ensure a quick and easy robot motion planning. The result of humanoid robot stability margin is 3.44 mm and its velocity 8.11cm/s.

Keywords: Stability, Zero Moment Point, , Invers Kinematic

I. Introduction

The development of technology has created the humanoid robot which is useful to help human daily activities [1]. This kind of robot has been investigated to improve its performance by many researchers. Humanoid robot can be defined as the robot that has similar appearance as human, however it is just consisted of the parts of human body system [4]. Commonly, humanoid robot has two legs as the foundation for its motion.

Humanoid robot is the closest robot that can replace human; however it has many defects especially its stability. Human depends on its stimulation and brains to create its stability system, and this kind of system is not existed in the robot. Robot needs the detail instruction and it must be programmed into the robot accurately [2]. Other obstacle that should be faced is the motion of robot

that will not be same as human respect to its different of flexibility. Trajectory control, balancing control, and motion planning are the three main controls that needed to make a balance motion in the humanoid robot [3]. Besides of these two obstacles, dynamic stability motion shall be investigated to obtain better motion in the humanoid robot.

Movement path of robot obviously will follow motion system of human. A step can be divided into some phases, but the simplest one is two phases which is Single Support Phase (SSP) and Double Support Phase (DSP). SSP is the motion of robot with one foot contact to the ground. On the other hand, DSP is a phase where the two foots of humanoid robot has a contact with the ground. SSP is divided into two other phases which are right SSP and left SSP. On one sequence of motion, the humanoid robot need four DSP motion and two SSP motion [6].

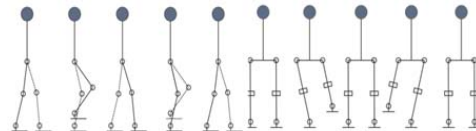


Fig.1 Walking sequence dari x-axis dan y-axis [9]

There are two balance states in the robot. The first one is static stability and the other one is dynamic stability. In the static stability, robot can move whenever it wants. On the other hand, dynamic stability needs to keep its balance state as it is moving [5]. A robot with one leg is lifted up need to focus its weight into one other leg in the static stability; however it will be different in the dynamic stability. As the robot is moving, it will ignore its static stability [1]. The utilization of Zero moment point in the dynamic stability analysis is the main point to investigate the motion of humanoid robot. This simple method has been applied by many researchers, though the location of Zero moment point need to be defined first.

The goal of this research is to approach humanoid robot stability motion to improve its motion like the human motion. The focus of this research is to design of robot with better stability and compare many materials in order to reach the best stability in the humanoid robot. Simulation is applied to this research as the comparison with the real motion of humanoid robot.

II. Methodology

2.1 Inverse Kinematic

In Kinematics, there are forward kinematics and inverse kinematics. The differences between them, forward kinematics will questioned position of the end effector when all of the angle in every joint which is servo is given. While, inverse kinematics answered the angle of servo when the position is given [7].

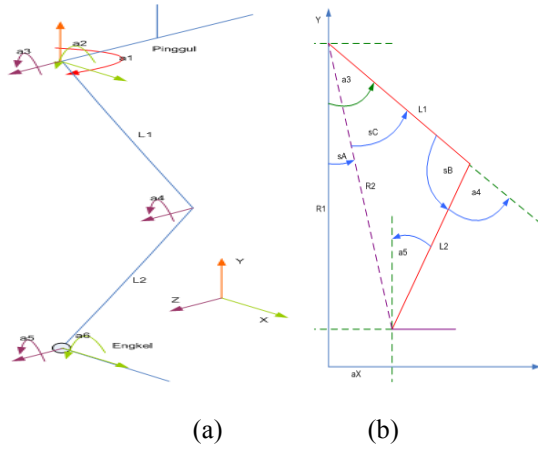


Fig.2 Inverse Kinematic Modelling on X-Axis Leg[7]

After doing the modeling of kinematics then we need to take from three perspectives, namely x-axis y-axis and z-axis. So first we will review the terms of the z axis is a side view of a robot. The concept of modeling of kinematics is shown in the above will get the values of the angle of each servo. If we want to rise the magnitude of footfall so that the parameters of his legs is high footfall is doing by inverse kinematics. By this method the position of high footfall will be adjusted by the servo-servo which form certain angles. It is known that tends thigh length $L1$ and $L2$ is the length of shank robot. The angle of the servo needed is $a1$, $a2$, $a3$, $a4$, $a5$ and $a6$ if it is to form a high footfall and large footsteps. With the approach of the triangle trigonometry cosine to obtain the required angles so the angle $a3$, $a4$ and $a5$ it first has to get the angle sC , sB and sA . With trigonometric equations then

$$L2^2 = L1^2 + R2^2 - 2 L1 R2 \cos sC \quad (1)$$

$$sC = \cos^{-1} \frac{L1^2 - L2^2 + R2^2}{2 L1 R2} \quad (2)$$

Afterwards, angle of sA is necessary known because $R1$ projection is needed to know in z-axis for calculation of another servo angle. Using Sinus law then :

$$\sin sA = \frac{aX}{R2} \quad (3)$$

$$sA = \sin^{-1} \frac{aX}{R2} \quad (4)$$

Because of sC equation is known so angle of sB will be compared to sC . Then, sC will be known :

$$\sin sB = \frac{R2 \sin sC}{L2} \quad (5)$$

$$sB = \sin^{-1} \left(\frac{R2 \sin sC}{L2} \right) \quad (6)$$

After that, all of the angle is known with the equations below:

$$a3 = sA + sC \quad (7)$$

$$a4 = 180 - sB \quad (8)$$

$$a5 = a4 - a3 \quad (9)$$

Three equations above can not be completed because $R1$ variable is still unknown. So it is necessary to know 3 others $a1$, $a2$, $a6$. Knowing that 3 equation must be referred from front look of robot.

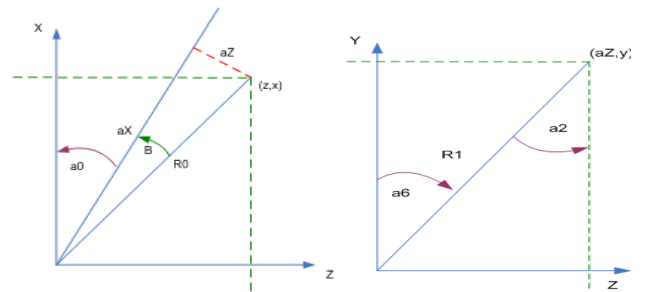


Fig.3 Robot in Y- Axis and X-Axis

Modeling on the coordinate axis Z towards X is analyzed first the $R0$. $R0$ is a projection of the shank length of the robot to the axis y-axis is a projection thigh while aX robot and vice versa. So the equation of the line projection obtained the following equation:

$$aZ = R0 \sin B \quad (10)$$

$$aX = R0 \cos B \quad (11)$$

Then after getting aZ , then it needs to review large servo angles $a2$ and $a6$ of the facade, x-axis, as shown. Modeling on the coordinate axis Z to Y is analyzed first the $R1$. This is a projection of $R1$ $R2$ at modeling z axis before. SA angle in the equation is helpful to know the magnitude of $R1$. First of all need to know the magnitude of $a2$ with the approach of a right triangle and the opposite corner is

$$\tan a2 = \frac{aZ}{Y} \quad (12)$$

$$a2 = \tan^{-1} \left(\frac{aZ}{Y} \right) \quad (13)$$

Then, $a2$ angle is opposite with $a6$ angle so

$$a2 = a6 \quad (14)$$

Kinematics concept that shown in modeling will get the angle of each servo when give the magnitude of foofall .

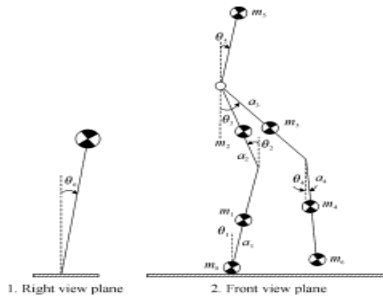


Fig.5 Parsial Centre of Mass [5]

In the image shown that the Centre of Mass (COM) on the robot is divided into several parts: thighs, calves and ankles as well as the body. Body is considered one just because essentially run more affected by foot even though the swing arm is also slightly affected. Then, the method of inverse kinematics forwarded to approach dynamic walking for more real way to approach the movement path of the robot.

2.2Zero Momen Point

After knowing of the robot kinematics and dynamics as well as the entire deployment force and pressure on the bracket robot then zero moment point method is used. This method analyzes the point where there is no moment.

$$Xzmp = Xcom - \frac{Zcom}{g} \ddot{X}com \quad (15)$$

$$Yzmp = Ycom - \frac{Zcom}{g} \ddot{Y}com \quad (16)$$

where $Xcom$, $Ycom$ and $ZCOM$ is the position of Centre of Mass of the robot as a whole, g is the acceleration due to gravity (m/s^2) and $\ddot{X}, \ddot{Y}$ is the acceleration in the x and y axis [1]. From the equation above, it is known that accelerated greatly affect the position shift. If the acceleration is considered 0 then it can be said that the required parameters are centre of mass position and is called static walking.

But this approach is less close to real because sometimes the inertial forces that affect the acceleration that made stable so that acceleration has a very important role. Results $Xzmp$ and $Yzmp$ must be in an area of the foot.

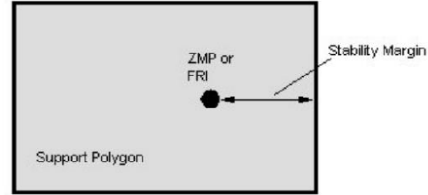


Fig.6 Stability Margin in SSP [7].

ZMP positions are reflected on a footstool at the time of DSP and SSP. As in the picture, magnitude of stability margin is distance between the ZMP point to nearest position on the outer line of support polygon. Getting closer to outer line means that it will be more unbalanced. Moreover, if the point ZMP on the robot are not in support polygon then it will fall.

III. Result and Discussion

The stability of robot can be measured using stability margin, however camera plotted point must be created with the recorded angle of walking cycle from the humanoid robot.



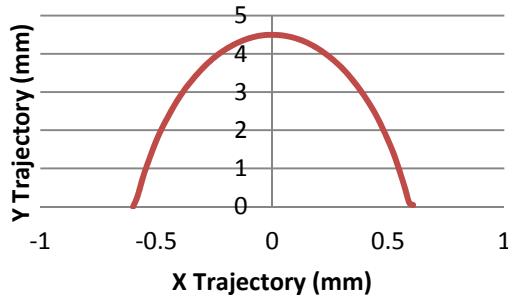


Fig.7 Trajectory of Z axis Robot Foot

The recorded angle was tested to obtain its foot height 45 mm, step forward 12 mm and Center of Mass of every single phase of humanoid robot motion path. Center of mass can be investigated using inventor simulation with the help of line in the inventor software. The beginning phase of walking motion is 8, 5.2, 24 cm for Xcom, Ycom, and Zcom respectively. This point was projected to ZMP using the equation, therefore Xzmp and Yzmp can be calculated which is 8 and 5.2 respectively. The distribution of ZMP can be seen in figure 7 with different acceleration.

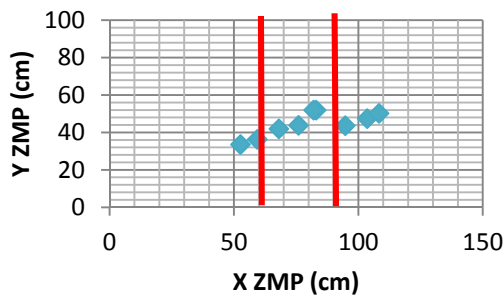


Fig.8 Distribution ZMP on foot sole of Robot

The distribution of ZMP points based on the foot sole can be described in the figure 8. The illustration is used to ease the calculation of the stability margin of humanoid robot. Based on the graph, the friction to the left direction from the right direction is displayed. The red line refers to the boundary between left foot and right foot. The middle margin is called y-offset or the margin between left foot and right foot. From the scatter diagram, there are many scatter points outside of the line. Inside of foot sole margin is SSP pose. On the other hand, outside of foot sole is DSP. Based on the data, it can be concluded that robot still has good balance and stability for walking activities.

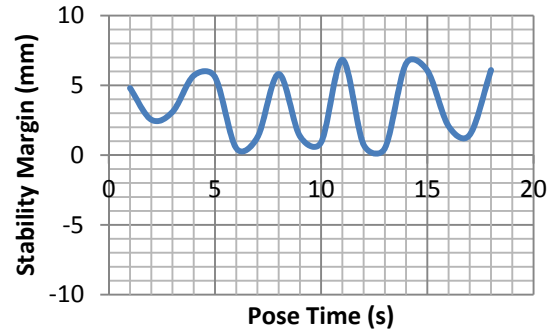


Fig.9 Graph of Stability Margin on DSP and SSP

From these two graphs are shown that ZMP points are still inside the support polygon which means robot is still in stable condition. Although, in some circumstances, ZMP points almost touch the outer line of support polygon and from the figure 11 shown that the steep gradient in some conditions but it is caused of transition phase from DSP to SSP. The magnitude of Stability Margin is positif which mean robot is in safe condition. The result of humanoid robot stability margin is 3.44 mm and its velocity 8.11cm/s.

Stability performance of humanoid robot can be measured by Global Centre of Mass Measurement. Following this equation from robosoccer competition rule :

$$L = \frac{(2.2 \times ZcoM)^2}{32} \quad (17)$$

where L is large of foot sole and ZcoM is centre of mass in Z Axis. From that equation, concluded that if amount of Zcom is higher so L is also bigger. The bigger amount of L affect the bigger amount of stability margin. So, Humanoid robot stability can be enhanced if ZcoM in higher position from ground. Compared to Zcom of DarwinOP which is 23 cm, this development robot Zcom is 24 cm. It can be achieved by selection of material on bracket.

IV. Conclusion

The points of this research conclusion are :

1. Designed robot stability has the average of stability margin which is 3.44 mm with the value of SSP 1.439 mm and DSP 5.93 mm
2. The velocity of Robot is 8.11 cm/s and acceleration of 0.551 cm/s²
3. Simulation using inventor can be developed as stability simulation software and velocity of humanoid robot
4. Stability performance of humanoid robot can be improved from Zcom robot 24 cm and Darwin OP 23 cm which mean its stability can be enhanced.

V. Reference

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